Paired Fault Assessment of a Fly-Connectome Spiking Controller

Shuchao Gao
Waseda University
Tokyo, Japan

***Abstract*—Task-level fault loss in a spiking controller can depend on feedback even when healthy success counts match. We provide an auditable paired assessment with shared input variates and deletion masks, causal feedback, exact scheduled actuation and independent input/geometry replay. A 138,639-neuron fly-connectome model is tested on 24 new seed clusters crossing two laws, two bearings and three network conditions. All 96 healthy episodes remain collision-free through one second. At 50% connection-row deletion, proportional and relay control succeed in 37/48 and 48/48 episodes; their paired loss difference is 22.9 percentage points, with a 95% cluster interval of 12.5–33.3 and Holm-adjusted $p = 0.00195$. The 25% contrast remains uncertain. A frozen 400-ms command-rate criterion detects only 2/13 later collisions. Geometry-informed forecasting detects all 13 but flags 57/275 successful tasks. Task loss changes with the declared horizon; stopped-duration matching leaves new group activity medians unchanged. These findings support paired task assessment without establishing biological or hardware fault tolerance.**

***Index Terms*—spiking neural networks, fault assessment, feedback control, connectome, neuromorphic systems**

I. Introduction

An embedded spiking controller must act before contact. A neural-rate criterion can accept a failed task, while altered activity can accompany success. Assessment requires a defined plant, causal actuation and a declared horizon.

Feedback couples faults to future sensory input. Equal seed labels need not align inputs when stopping or event consumption changes. Early termination also changes accumulated spikes: stopped, common-duration and fixed-horizon counts answer different resource questions. Pairing requires verified healthy capability in every direction.

We couple a fly-connectome SNN to synthetic pursuit avoidance. Assumed receptive fields and body coupling define an engineering stress model, not biological escape. Separate development and validation establish bilateral laws; new seed clusters cross law, direction and nested deletion dose at one second.

Our contribution is an executable paired protocol with source-bound registrations, shared input variates and masks, exact actuation and independent input/geometry replay. Prospective testing demonstrates feedback-dependent fault loss despite identical healthy success counts, and quantifies neural warning errors. Healthy reserves and actions differ: this is a whole-system comparison. Standard statistics and kinematics support bounded findings, not algorithmic novelty. Failed development and all test cells are retained.

II. Positioning and Scope

Shiu et al. [1] connect anatomical wiring to executable sensorimotor hypotheses. An's sparsification study [2] varies connection strength, placement, state and task; Jo [3] reports neural and virtual-arena responses to connectome damage. These works already establish context- and endpoint-dependent function. Our question concerns source-auditable pairing, sampled-data actuation and same-record exposure estimands in a prospective assessment, rather than a new claim that endpoint choice matters.

SpikeFI [4] and SpikingJET [5] provide SNN injection frameworks with application evaluation. NeuroBench [6] and embodied-agent benchmarking [7] motivate separate functional and resource measurements. Here the comparisons rescore our own generated records; neither injector framework is executed as a benchmark. Physical neutron tests [8] illustrate the additional evidence needed to connect software fault models to hardware. Our weighted-row deletions have no calibrated physical upset rate, power or area interpretation.

III. Paired Closed-Loop Assessment Protocol

A. Substrate and Randomness

The released FlyWire v783 graph has 138,639 neurons and 15,091,983 signed connection-table rows, each representing multiple contacts. We use the Shiu LIF parameterization [1] in Brian 2 [9], NumPy backend, 0.1 ms steps. The original publication used v630. Sensory cells retain 2.2 ms refractoriness, a stated stimulation variant. Initial states and unmodified weights are restored before each episode; equations and parameters accompany the source.

An assumed receptive-field map supplies geometry-dependent rates up to 150 Hz to 104 LC4 and 210 LPLC2 cells. Other external rates are zero. Eight descending cells form a synthetic command interface. Anatomy and looming selectivity [10] motivate the substrate; the map and body coupling are not validated fly behavior. Actual cell identities and map parameters are saved.

Each seed crosses both bearings, both laws and healthy/25%/50% deletion. Shared row variates define $w_j^{(f)} = w_j I(V_{sj} \geq f)$, giving nested masks. All contacts in a row are removed together. Separate frame-indexed uniforms generate inputs $I[U_{sk\tau i} < \lambda_{ki}\Delta t]$. Uniforms are common across conditions; realized events differ through feedback-dependent λ. Digests bind masks, uniforms, actual

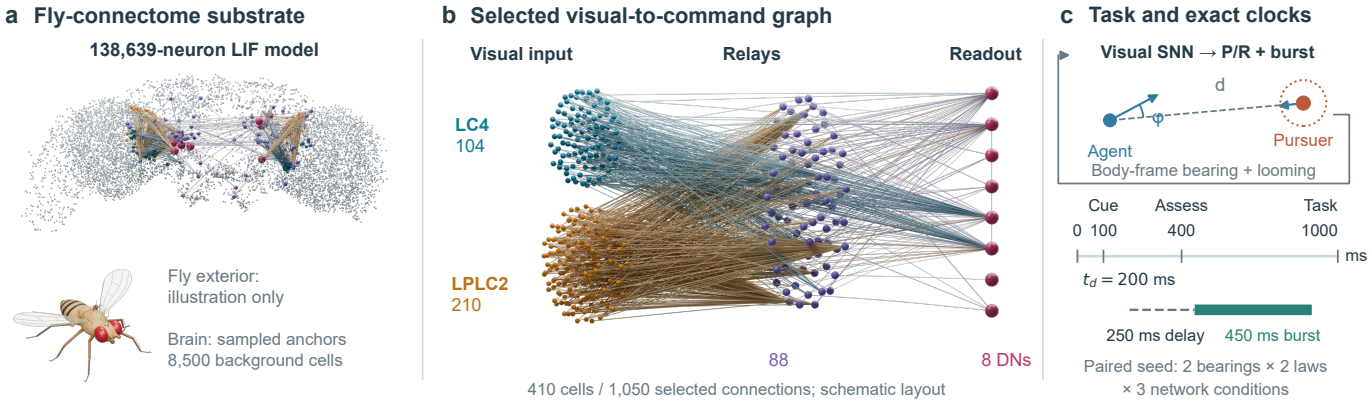


Fig. 1. Model-to-task interface. (a) Representative anchors and a separate illustrative fly. (b) A 410-cell/1,050-connection subgraph, with schematic layout and routes, not measured neurites or the full recurrent graph. Colors denote groups. (c) Synthetic task, next-frame steering and exact burst timing for $t_d = 200$ ms. P/R: proportional/relay control.

events, maps, sources and external tables. Seed traceability does not imply cross-platform bit identity.

B. Causal Feedback and Actuation

Body-frame bearing subtracts heading from world-relative bearing. Looming uses changes in $\alpha = \arctan(0.05/d)$ at actual separation d. Input begins at 100 ms; decisions at each frame end affect the *next* 20 ms motion interval.

Twenty passive episodes (four seeds; $-60, -30, 0, 30, 60°$) fit a ridge decoder $z_k = \beta_0 + \boldsymbol{\beta}^\top \bar{\mathbf{r}}_k$ of bearing/60°, using five-frame means of eight command rates. Both laws set $\omega_k = 0$ before the cue or below the 4.03125 Hz eight-cell mean trigger level; otherwise

$$\omega_k = \begin{cases} -600\,\mathrm{clip}(z_k, -1, 1), & \text{proportional}, \\ -600\,\mathrm{sgn}(z_k), & \text{relay}, \end{cases} \quad [\mathrm{deg/s}]. \quad (1)$$

The first qualifying decision t_d schedules one 0.35 m/s heading-fixed burst on $[t_d + 250, t_d + 700)$ ms. Otherwise speed is 0.05 m/s and heading is capped at $\pm 100°$. Motion segments split at exact actuator boundaries; positions use starting heading, updated at each segment end. A 0.3 m/s pursuer starts 0.25 m away and aims at the agent at each segment's beginning. First contact at distance $\leq$ 0.05 m is solved inside each linear relative-motion segment.

Success means collision-free through 1 s, not eventual escape. Walk-only segments satisfy $d_{\mathrm{next}} \leq d - 0.25\delta$ (δ in seconds), so contact occurs by 0.8 s without a burst in this plant. Failure latches at first contact; the trajectory and neural loop continue for equal 1 s resource exposure. Post-contact activity is counterfactual accounting, not physical collision dynamics or warning data. Independent replay checks inputs, geometry, decisions, clocks and horizon state without rerunning neural dynamics.

C. Paired Estimands

Let $S_F(t), S_H(t)$ be paired network counts, and T_F, T_H their contact-frame end or 1 s. With $T_* = \min(T_F, T_H)$, distinguish

$$R_{\mathrm{stop}} = \frac{S_F(T_F)}{S_H(T_H)}, \quad R_{\mathrm{common}} = \frac{S_F(T_*)}{S_H(T_*)}. \quad (2)$$

Fixed 400 ms and 1 s ratios are separate estimands. The former precedes all contacts; common duration remains outcome-dependent and does not equate sensory histories. For binary outcomes, report loss $Y_H(1 - Y_F)$, gain $(1 - Y_H)Y_F$ and net loss $D = Y_H - Y_F$. The controller interaction is $D_{\mathrm{relay}} - D_{\mathrm{proportional}}$, averaging bearings within seed.

D. Validation, Warnings and Inference

An initial controller failed its declared capability screen; all attempts are retained. Final calibration passed on eight new healthy validation clusters, 8/8 per law/bearing. A two-cluster fault screen was developmental. Core testing uses 24 new clusters (288 episodes); eight separate $\pm 45°$ clusters test angle transfer. Parameters, thresholds and stopping rules were frozen before both tests.

Warnings use only data through 400 ms: low mean command rate, an upper activity limit, and $|\log(S/\tilde{S}_H)|$ against law/side-specific healthy medians. A held-command forecast uses current geometry, neural turn command and pending burst; absent a trigger it assumes the earliest next-frame decision. A same-state actuator baseline sets future turn to zero. Both have exact simulated geometry, not implemented sensors or reachability guarantees. Lower fifth-percentile command/clearance and upper 95th-percentile log-score thresholds use all healthy validation records. Cutoffs are 5.833 Hz (100–400 ms command mean), activity ratios outside $[0.844, 1.185]$, and predicted clearance below 18.77/18.55 mm for held-command/schedule forecasts. The upper resource limit is $2\tilde{S}_H$. Healthy flag calibration does not guarantee task false-alarm rates.

All test records contribute sensitivity, specificity, false alarms and misses against contact through 1 s. Confidence intervals use 20,000 whole-seed percentile resamples, RNG 20261022. Controller sign-flip sensitivity assumes a symmetric cluster null, Holm-adjusted over two doses. Fixed

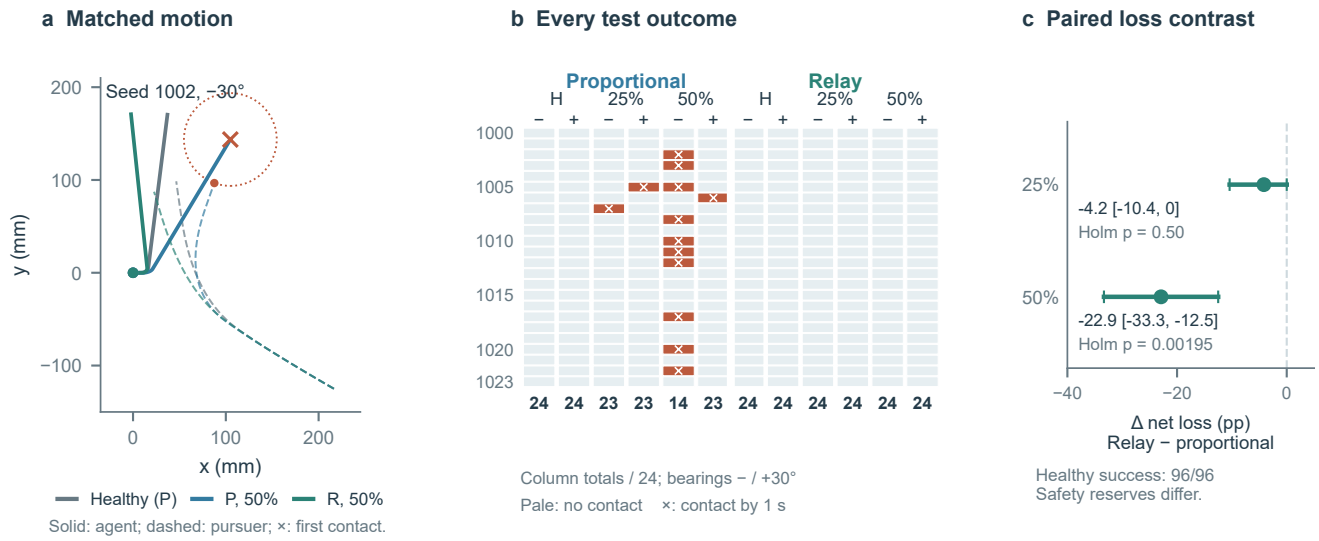


Fig. 2. Feedback and task loss. (a) Actual matched paths for seed 1002, $-30°$, the first seed-ordered proportional 50%-deletion collision, shown descriptively. Solid/dashed: agent/pursuer; paths stop at first contact, with 50-mm dotted contact radius. (b) All 288 outcomes in 24 seed rows; column totals are successes/24. Bearings, laws and doses within each seed are paired. (c) Registered relay-minus-proportional net-loss contrasts, 95% whole-seed bootstrap intervals and Holm-adjusted p. Healthy safety reserves differ.

800/900/1000 ms task horizons are repeated observations, not new replicates. No test tuning, seed replacement or efficacy stopping is permitted. Inference remains conditional on one model, map and task.

IV. Independent Test Results

A. Fault Loss Depends on Feedback

All 96 healthy core episodes succeed: 24/24 for each law/bearing in Fig. 2. At 25% deletion, proportional control succeeds in 46/48 episodes and relay in 48/48; at 50%, these counts are 37/48 and 48/48. There are no healthy-to-fault gains. The 50% proportional losses are direction-dependent: 10/24 at $-30°$, 1/24 at $+30°$. Both healthy directions remain capable, avoiding the earlier directional floor.

The registered relay-minus-proportional net-loss contrast at 50% is -22.9 percentage points (pp), 95% cluster interval $[-33.3, -12.5]$, with 11/24 nonzero cluster contrasts. Symmetric-null sign-flip sensitivity gives raw $p = 0.000977$, Holm-adjusted $p = 0.00195$. At 25%, the contrast is -4.2 pp, interval $[-10.4, 0]$, adjusted $p = 0.50$; two nonzero clusters provide limited evidence. Relay's zero losses are observations, not zero-risk estimates. These new laws do not replicate the older six-cluster original/gated comparison, whose exploratory 25% $p = 0.25$ remains in the supplement.

A post-hoc trace diagnostic finds identical trigger times and pre-decision traces in all 144 matched law/scenario pairs. Steering is gated below the trigger threshold, so the laws first diverge after that decision. Median trigger time rises from 160 to 220 ms under 50% deletion for both laws; median absolute heading at precontact 400 ms is $38.7°$ for proportional control and $96.0°$ for relay. Healthy median minimum clearance is 33.1 versus 41.4 mm. Thus identical success counts do not match safety reserve or executed turning effort. This describes a timing/turning pathway, not mediation or fault-specific superiority independent of baseline margin.

B. Neural Criteria Miss Task Failures

All 400-ms assessments precede the earliest contact, 607.6 ms. Fig. 3 scores the same 288 records against 13 later collisions and 275 successful tasks. The command-rate criterion detects only 2/13 collisions; its sensitivity interval is $[0, 40.0]\%$. Two-sided activity detects 3/13 with more false alarms, and the upper activity limit detects none. At this frozen calibration, adding a lower activity cutoff still misses 10/13 collisions.

Both geometry-informed forecasts detect 13/13 collisions. Holding the current neural turn command reduces false alarms from 84 to 57 relative to the same-state zero-future-turn baseline. A secondary descriptive paired analysis gives a specificity gain of 9.8 pp, interval $[6.6, 13.4]$. Its remaining 20.7% false-alarm rate and exact-state information limit practical interpretation. Zero observed misses yield a degenerate bootstrap interval; the one-sided 95% bound on any miss in a new, identically defined seed cluster is 11.7%, not a conditional FNR guarantee. This candidate is not a validated protection algorithm.

C. Prespecified Angle Transfer

Eight new $\pm 45°$ seed clusters retain the frozen map, laws and thresholds. All 32 healthy episodes and all 32 25%-deletion episodes succeed. At 50%, proportional control succeeds in 14/16 and relay in 16/16; both losses occur at negative bearing. The relay-minus-proportional contrast is -12.5 pp, interval $[-31.3, 0]$. This small stress cohort is not pooled with the core and does not establish a precise replication. Of two collisions, command rate flags one, two-sided activity none,

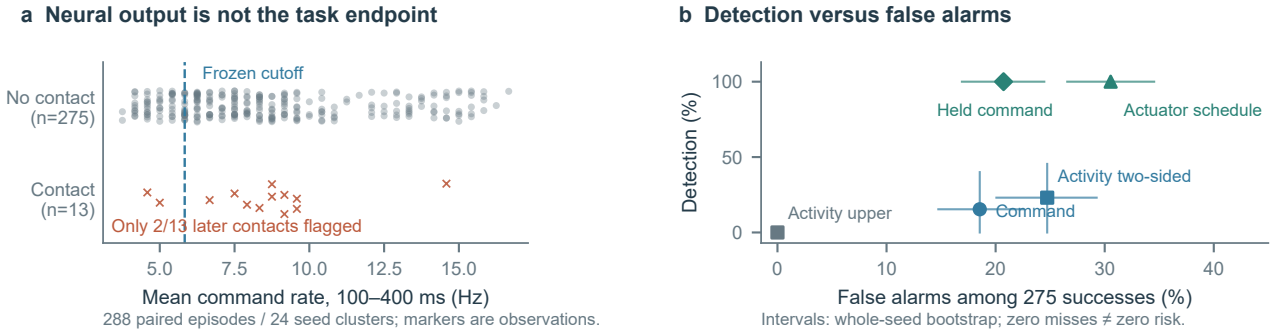


Fig. 3. Frozen 400-ms warnings on all core records. (a) Every command-rate observation with visual vertical jitter and the healthy-validation cutoff. (b) Detection among 13 collisions versus false alarms among 275 successes; 95% whole-seed bootstrap intervals. Geometric forecasts use exact state; zero misses do not establish zero risk. Complete confusion counts are in the supplement.

and both forecasts two; held-command/schedule false alarms are 11/94 and 21/94 successful tasks. With only two failure clusters, warning sensitivity remains poorly constrained.

D. Separate Action-Interface Check

An additional eight-seed diagnostic was registered after core launch and remains separate from its inference. On 64 new healthy-network episodes, normal proportional/relay control each succeeds in 16/16; turning disabled with burst triggering retained succeeds in 0/16, and both neural actions disabled in 0/16. Baseline walking remains active. These controls support action dependence in this plant; they do not establish connectome-specific superiority or separately isolate burst effects. The walk-only feasibility bound in Methods explains why the latter failure is partly constructed.

E. Horizon and Exposure Have Distinct Effects

Only two collisions occur by 800 or 900 ms, versus 13 by 1 s. The 50% controller contrast is therefore -4.2 pp at the earlier horizons, versus -22.9 pp at 1 s. These are repeated observations, not extra samples; the declared horizon materially changes task loss.

In contrast, duration matching leaves every group median spike ratio unchanged. At 50% deletion, stopped/common medians are 1.505 for proportional control and 1.257 for relay; every pair changes by less than 0.00325. The large historical stopped/common difference does not recur here. Proportional 50% activity has median ratio 0.981 at 400 ms and 1.505 over 1 s, illustrating distinct resource windows. Full-horizon post-contact continuation is not prospective warning evidence.

V. Executable Evidence and Audit Cost

One JSON file per paired condition records frames and actuator segments. Source, map, calibration and input bindings reject incomplete or mixed matrices. Independent replay reconstructs geometry, decisions and inputs without the task implementation or connectome; arithmetic checks verify effects and warning denominators. Mutation tests reject inconsistent headings, bursts and horizon flags and future-data leakage in prefix metrics. A tiny Brian 2 test checks state and fault-weight restoration.

The core neural campaign takes 14,393 s with three CPU workers; median episode wall time is 145.5 s. Independent trace/input replay takes 34.4 s. These perform different work: replay consumes recorded neural outputs and does not certify neural-state correctness or establish a simulation speedup. No hardware energy or area is inferred.

The package supplies raw JSON/CSV, generating snapshots, local registrations, TeX, figures and checksums; external tables remain separate. Registrations lack external timestamps. A logged CSV fix separates numeric-margin and Boolean-warning columns; frozen sources and raw records remain intact. Reanalysis/replay needs no neural execution. Public main may predate this revision.

VI. Implications and Limits

The same functional deletion mask can receive different task-level fault labels under two healthy-capable feedback laws. A neural threshold and a task requirement should therefore be reported separately, with actuator schedule, horizon and healthy safety reserve attached. At the tested calibration, a two-sided activity control also leaves substantial misses.

Evidence remains conditional on one v783 graph, one assumed visual map and a simple finite-horizon pursuit task. Healthy safety margins and turning effort are not matched; the law contrast does not isolate fault-specific tolerance from nominal reserve. The fixed rules are not optimized classifiers; the twofold spike envelope is not an established resource budget. Forecasts use more information than neural counts and retain substantial false alarms. Sparse failure clusters and fixed calibration limit uncertainty estimates. Row deletion is uncalibrated to physical bit errors; energy, area, real-time execution and biological tolerance are not established.

VII. Conclusion

Paired assessment separates feedback-dependent task loss from neural-rate and activity criteria. Independent testing supports the 50% deletion contrast at one second; the 25% contrast remains uncertain and stopped-duration matching changes little in this cohort. The reusable source-bound workflow specifies task, actuation and resource endpoints, with independent replay.

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